Lesson 2

GRAPHICAL PRESENTATION OF THE DATA

Excel in Business

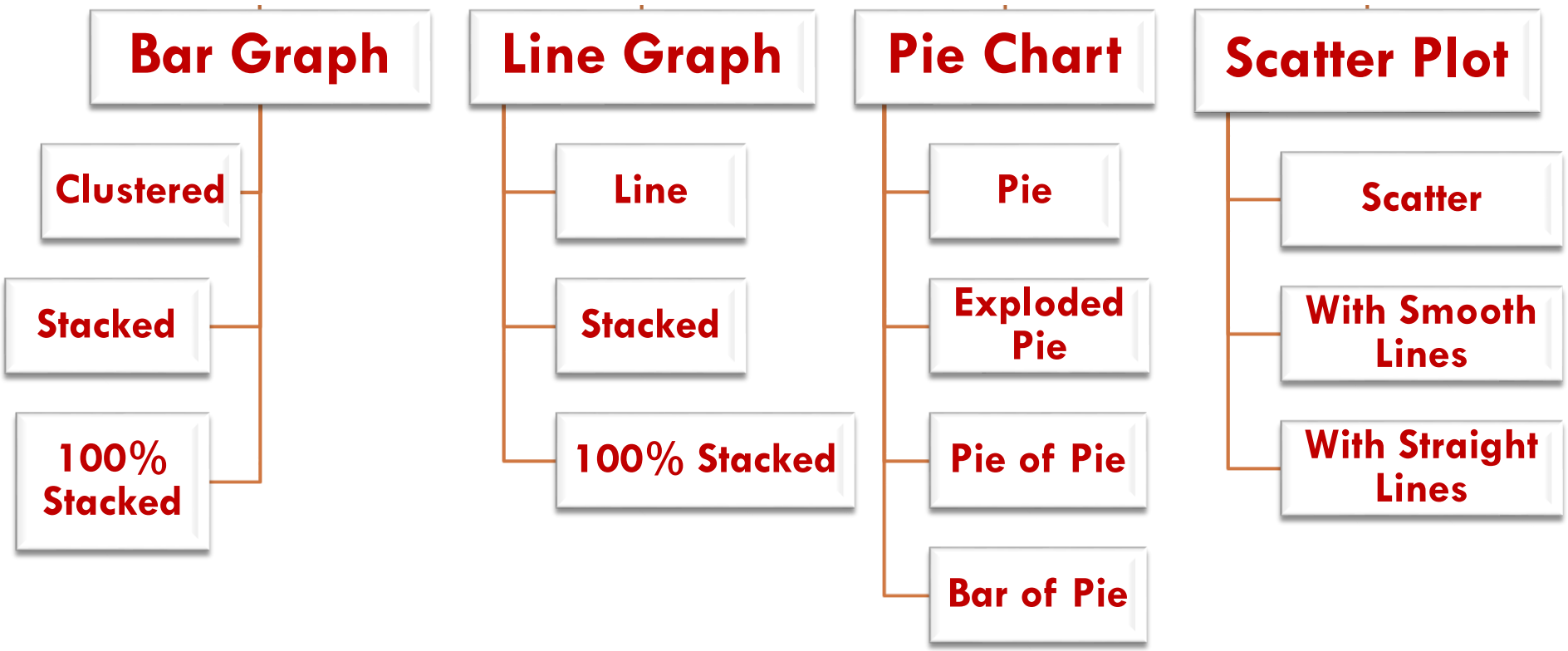
Importance

- Graphs give an attractive and elegant presentation. The significance of presented data is known without loss of time.

- With the help of graphs, comparisons of groups and series of figures can be made easily.

- One can study the trend and fluctuations.

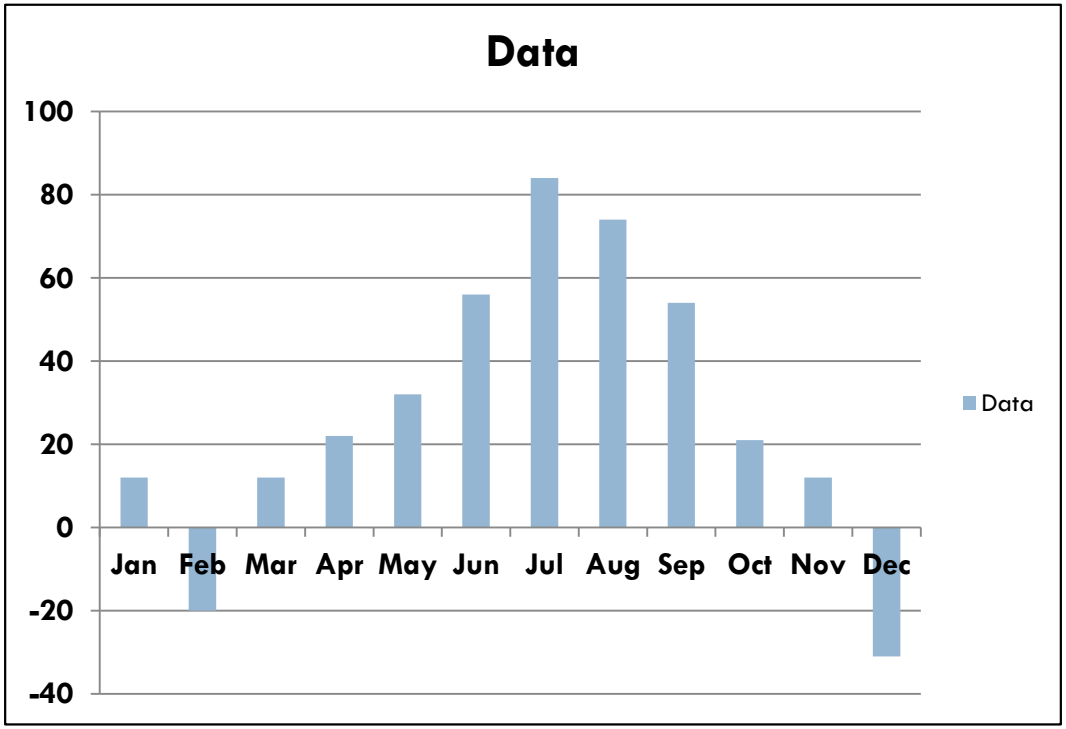
Available chart types in Excel 2007


Bar Graph
Clustered
Stacked
100% Stacked
Line Graph
Line
Stacked
100% Stacked
Pie Chart
Pie
Exploded Pie
Pie of Pie
Bar of Pie
Scatter Plot
Scatter
With Smooth Lines
With Straight Lines

Example 1:

The table below shows monthly data of some quantity.

Month	Data
Jan	12
Feb	-20
Mar	12
Apr	22
May	32
Jun	56
Jul	84
Aug	74
Sep	54
Oct	21
Nov	12
Dec	-31



Bar Graphs

A bar chart displays categorical data using a number of rectangles/bars, of the same width, each of which represents a particular category.

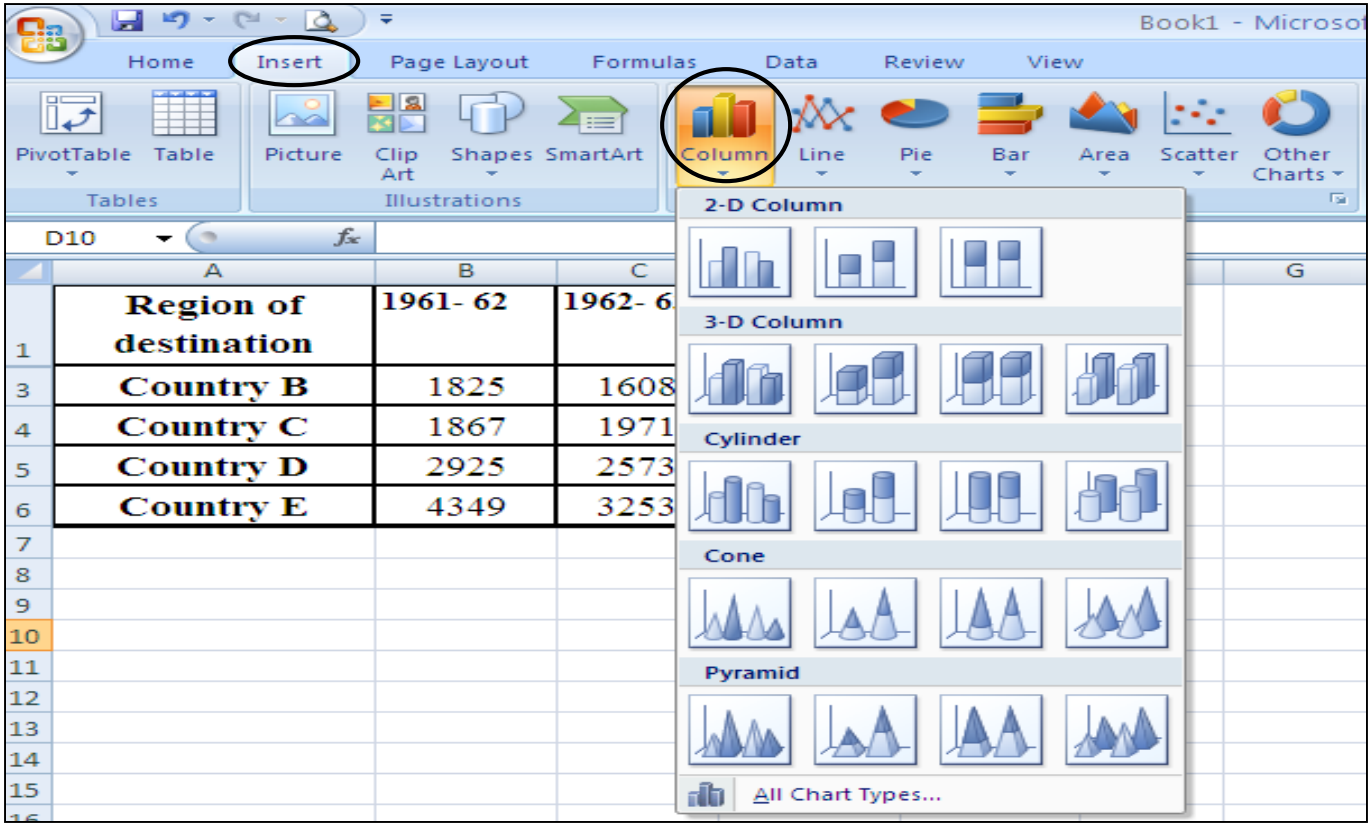
The length of each rectangle is proportional to the number of cases in the category it represents.

Rectangles can be drawn vertically or horizontally according to the preference and convenience.

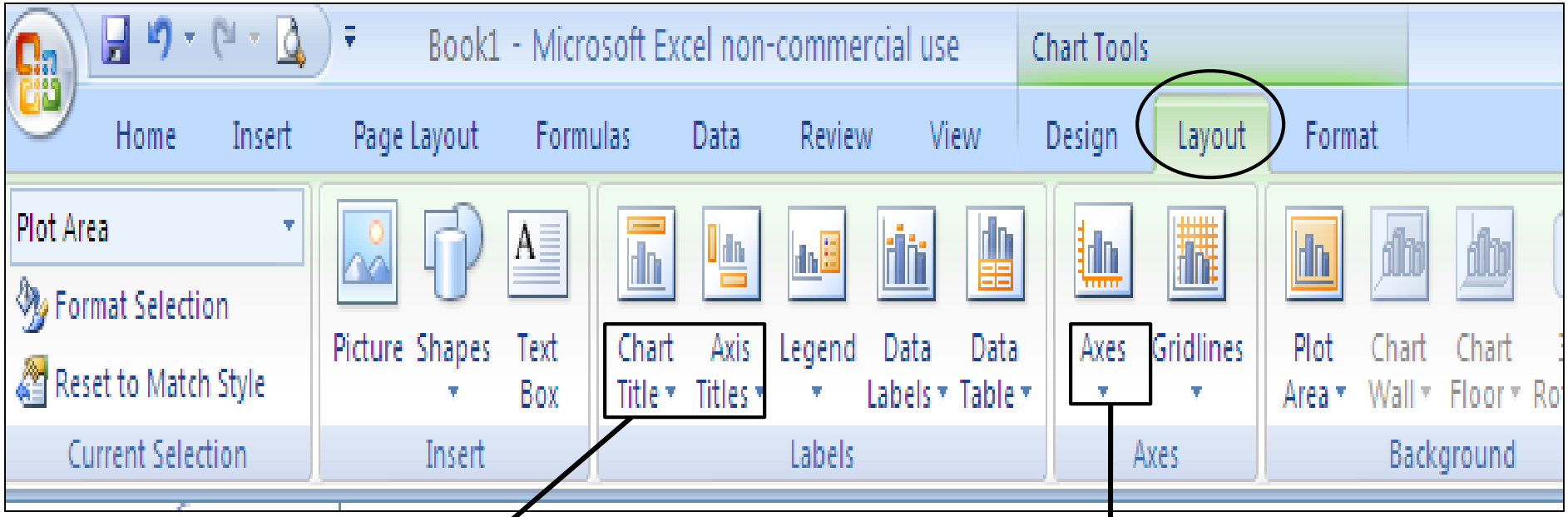
How to create a bar chart in Excel?

1. Select the data

2. Insert → Column/Bar → Select the appropriate chart- Clustered/ Stacked/ 100% stacked



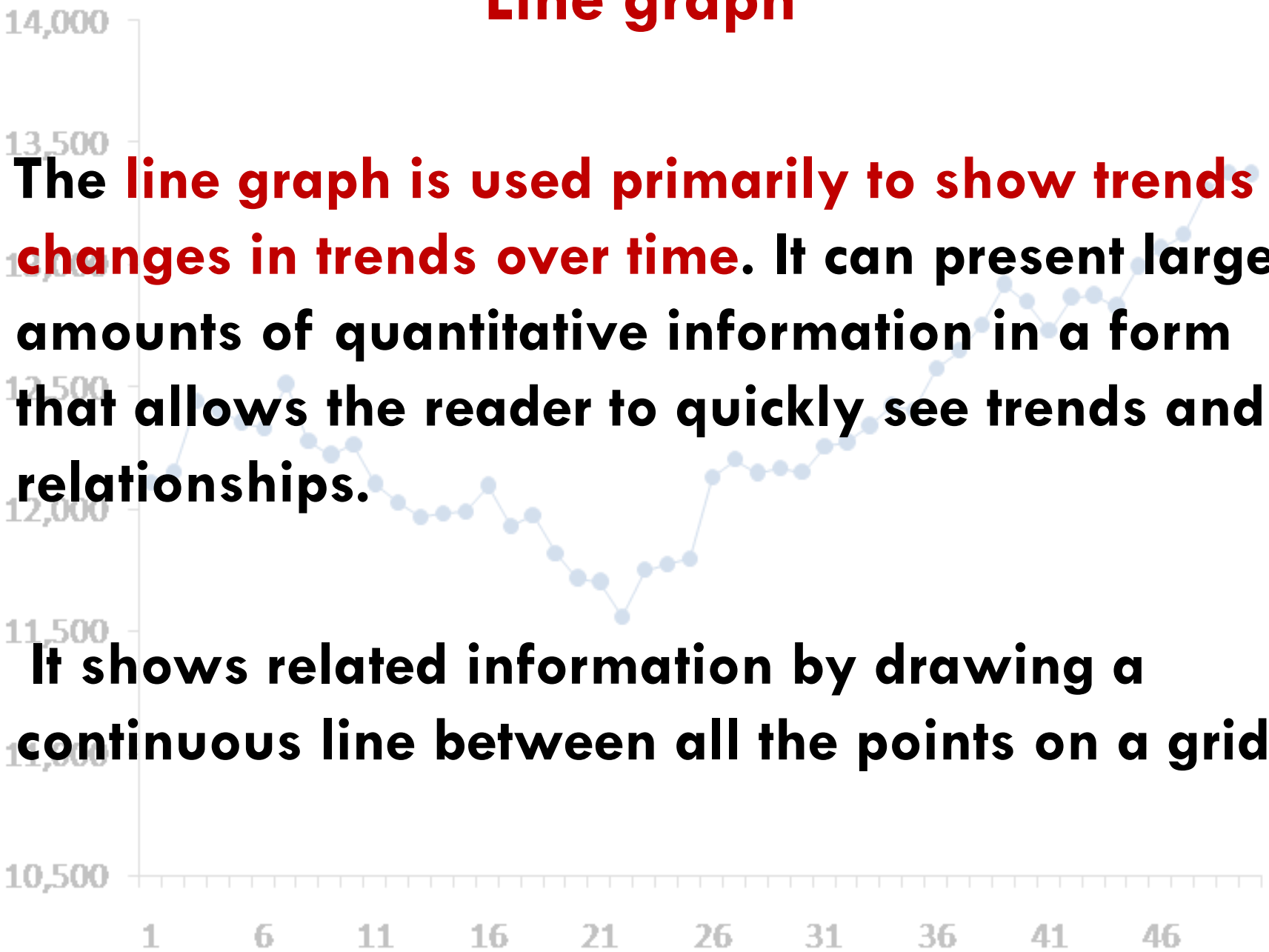
Formatting charts in Excel



Add title to chart and axes

Format axes

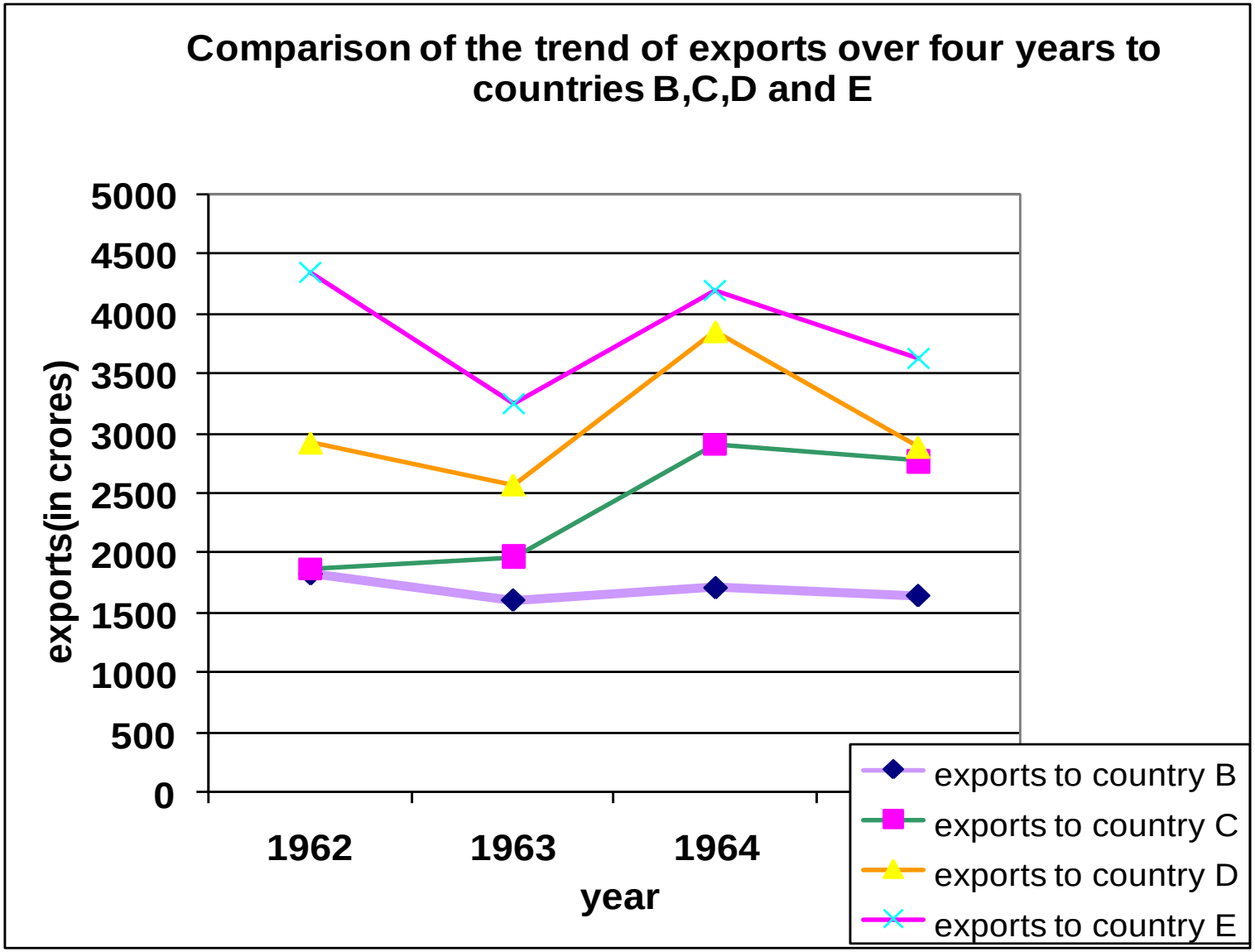
Line graph



The **line graph is used primarily to show trends or changes in trends over time**. It can present large amounts of quantitative information in a form that allows the reader to quickly see trends and relationships.

It shows related information by drawing a continuous line between all the points on a grid.

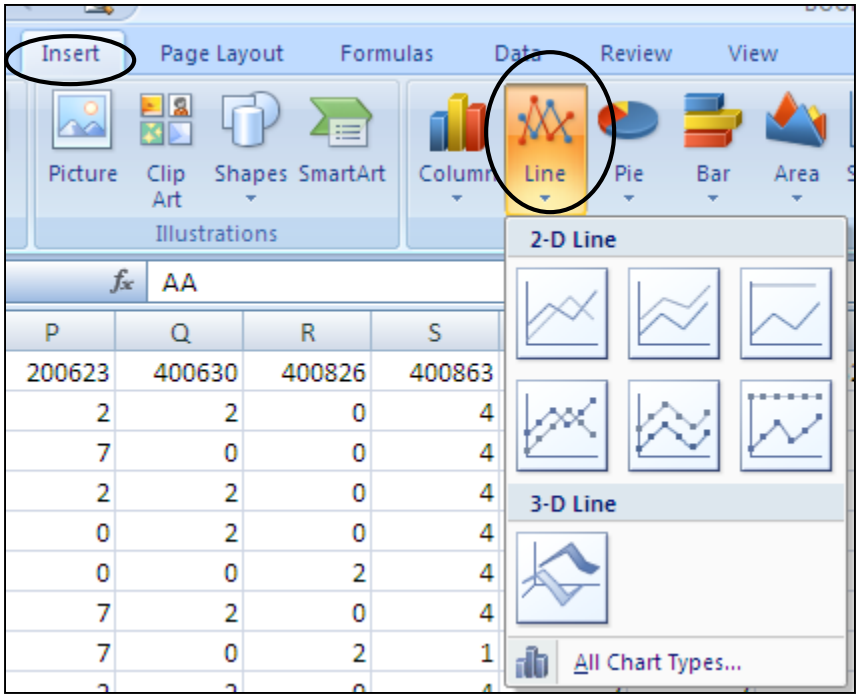
Example of a line chart



How to create a line chart in Excel?

1. Select the data

2. Insert → Line → Select the appropriate chart- Line/ Stacked/ 100% stacked with or without markers.



Note:

To avoid potential misinterpretation, the number of plotted lines in a single graph should not exceed five and normally should be limited to three.

Pie Chart

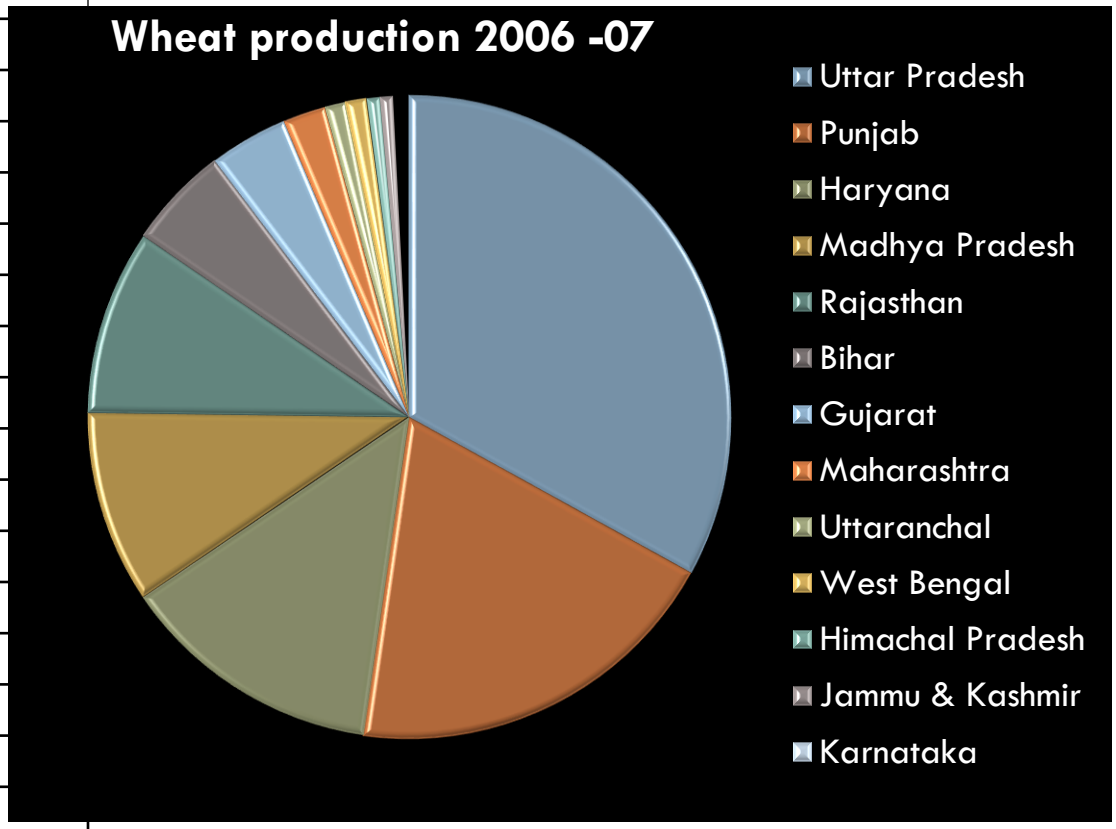
A pie-chart consists of a circle divided into sectors, each of which contains one category of information.

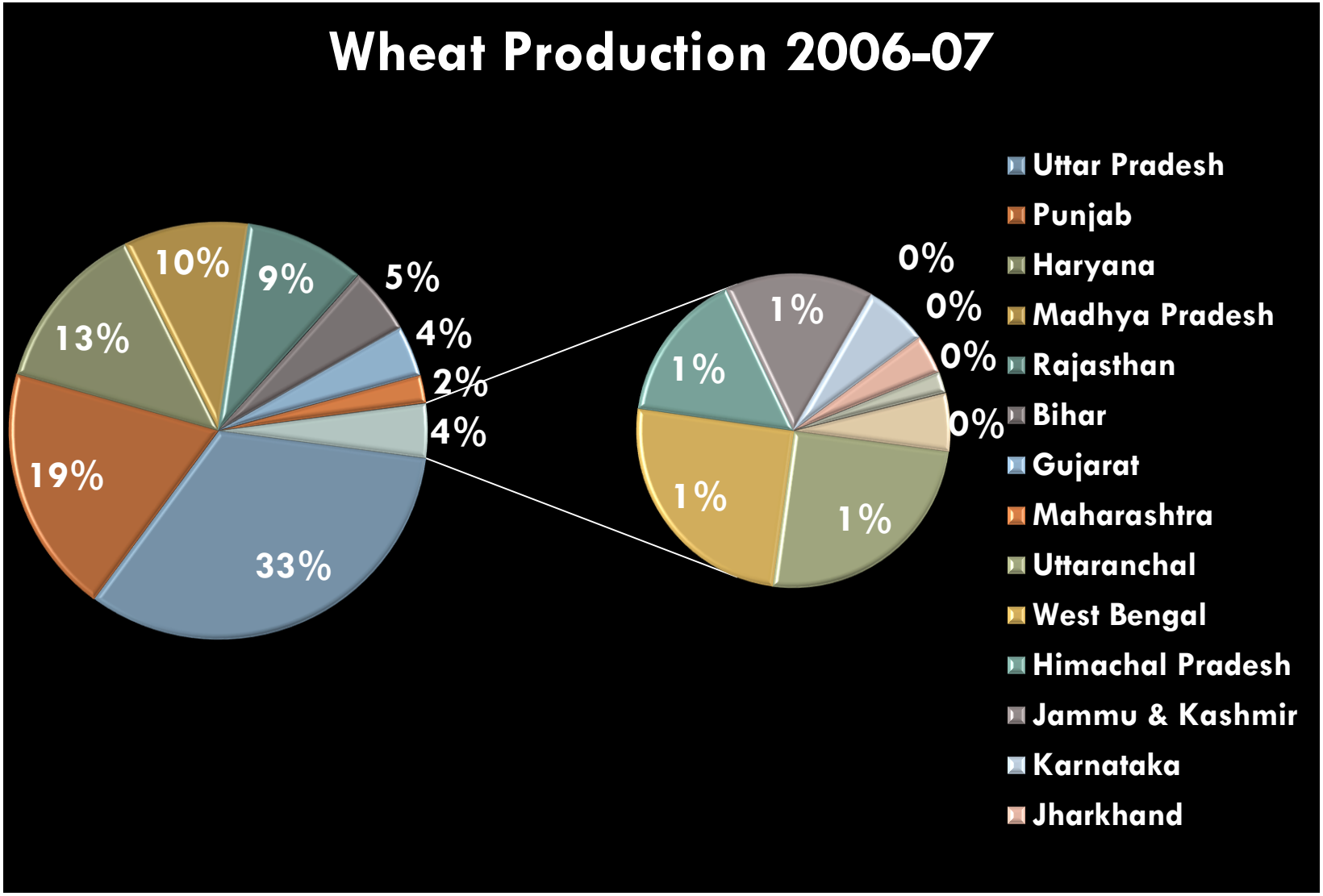
The area of each sector indicates the category's relative importance in relation to the other categories.

The circle of a pie graph represents 100%. Each portion that takes up space within the circle stands for a part of that 100%.

Example:

State	Wheat Production (Million tonnes) 2006-2007
Uttar Pradesh	25.03
Punjab	14.60
Haryana	10.06
Madhya Pradesh	7.33
Rajasthan	7.06
Bihar	3.91
Gujarat	3.00
Maharashtra	1.63
Uttaranchal	0.80
West Bengal	0.80
Himachal Pradesh	0.50
Jammu & Kashmir	0.49
Karnataka	0.21
Jharkhand	0.13
Assam	0.07
Others	0.19
All India	75.81

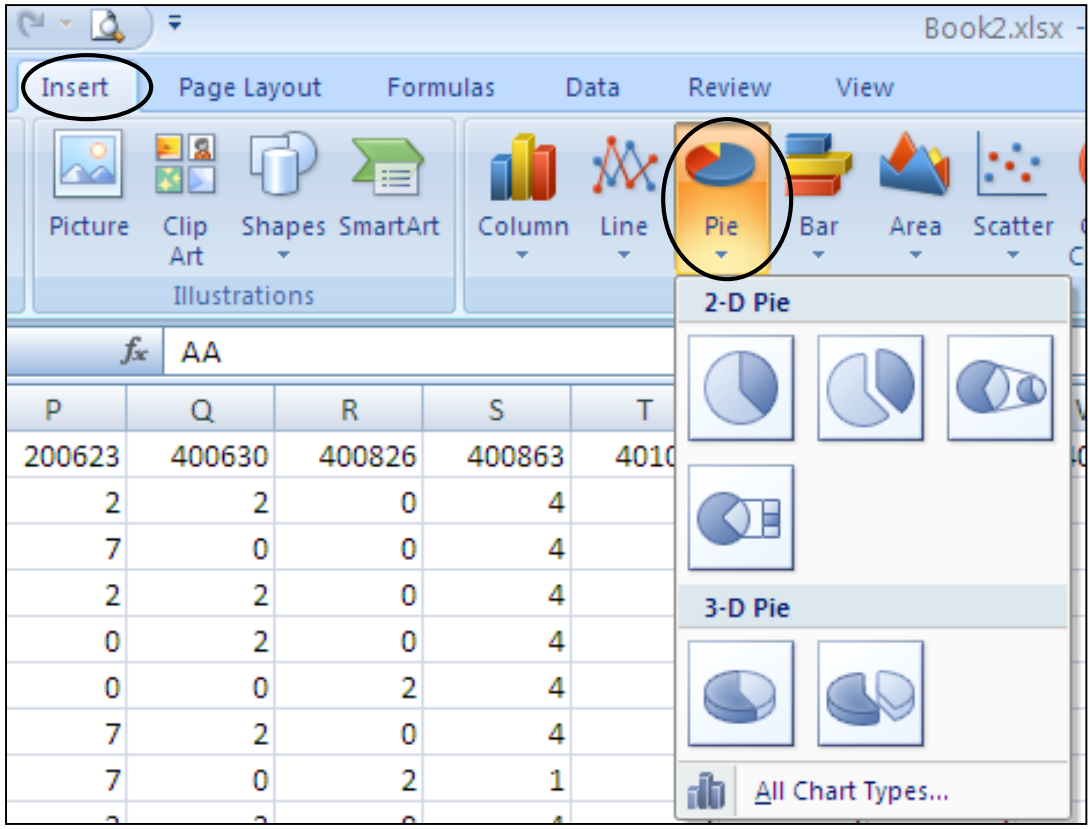



Wheat Production 2006-07
10%
9%
5%
4%
2%
4%
13%
19%
33%
0%
0%
0%
0%
1%
1%
1%
1%
Uttar Pradesh
Punjab
Haryana
Madhya Pradesh
Rajasthan
Bihar
Gujarat
Maharashtra
Uttaranchal
West Bengal
Himachal Pradesh
Jammu & Kashmir
Karnataka
Jharkhand

How to plot a Pie Chart in Excel

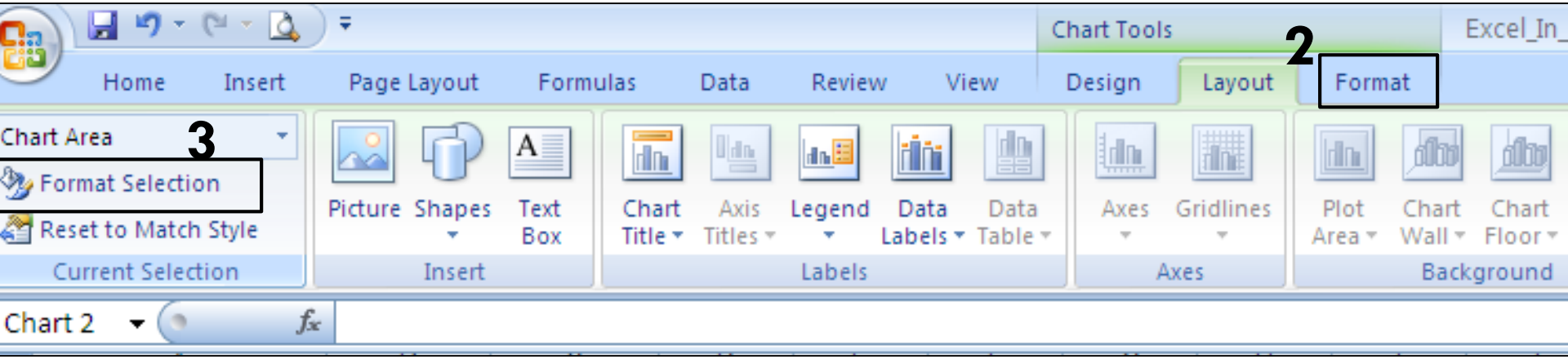
1. Select the data

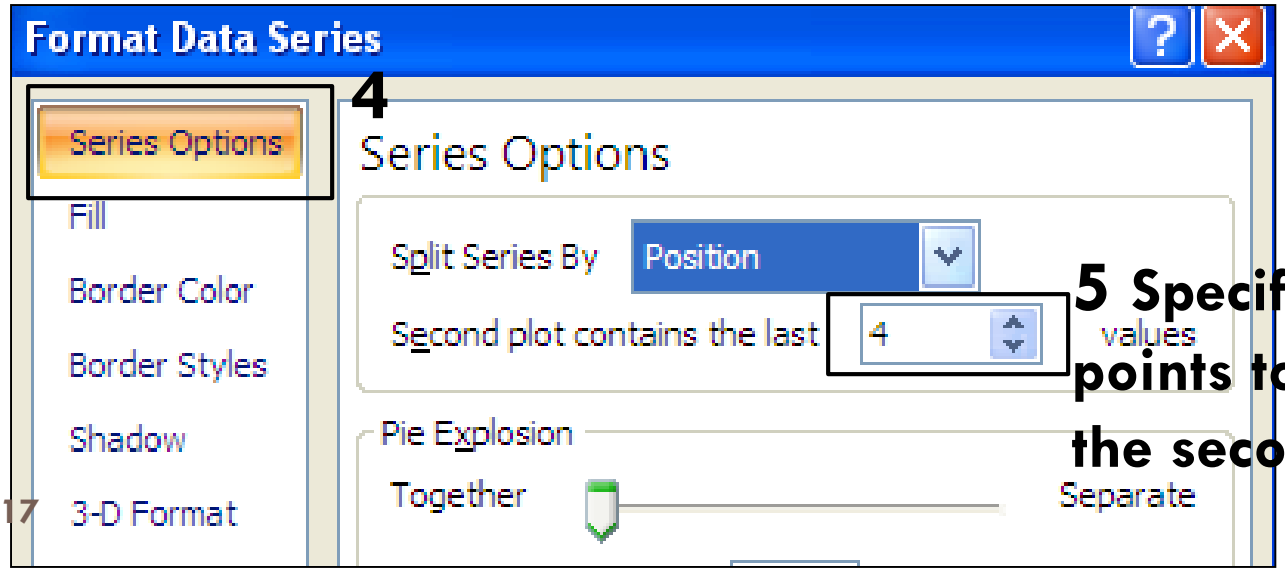
2. Insert → Pie → Select the appropriate chart- Pie/ Pie of Pie/ Bar of Pie.



To specify the number data points to be displayed in the secondary pie chart in case of a pie of pie chart:

1. Click on the secondary pie chart





5 Specify the number data points to be displayed in the secondary pie chart

Scatter Plot

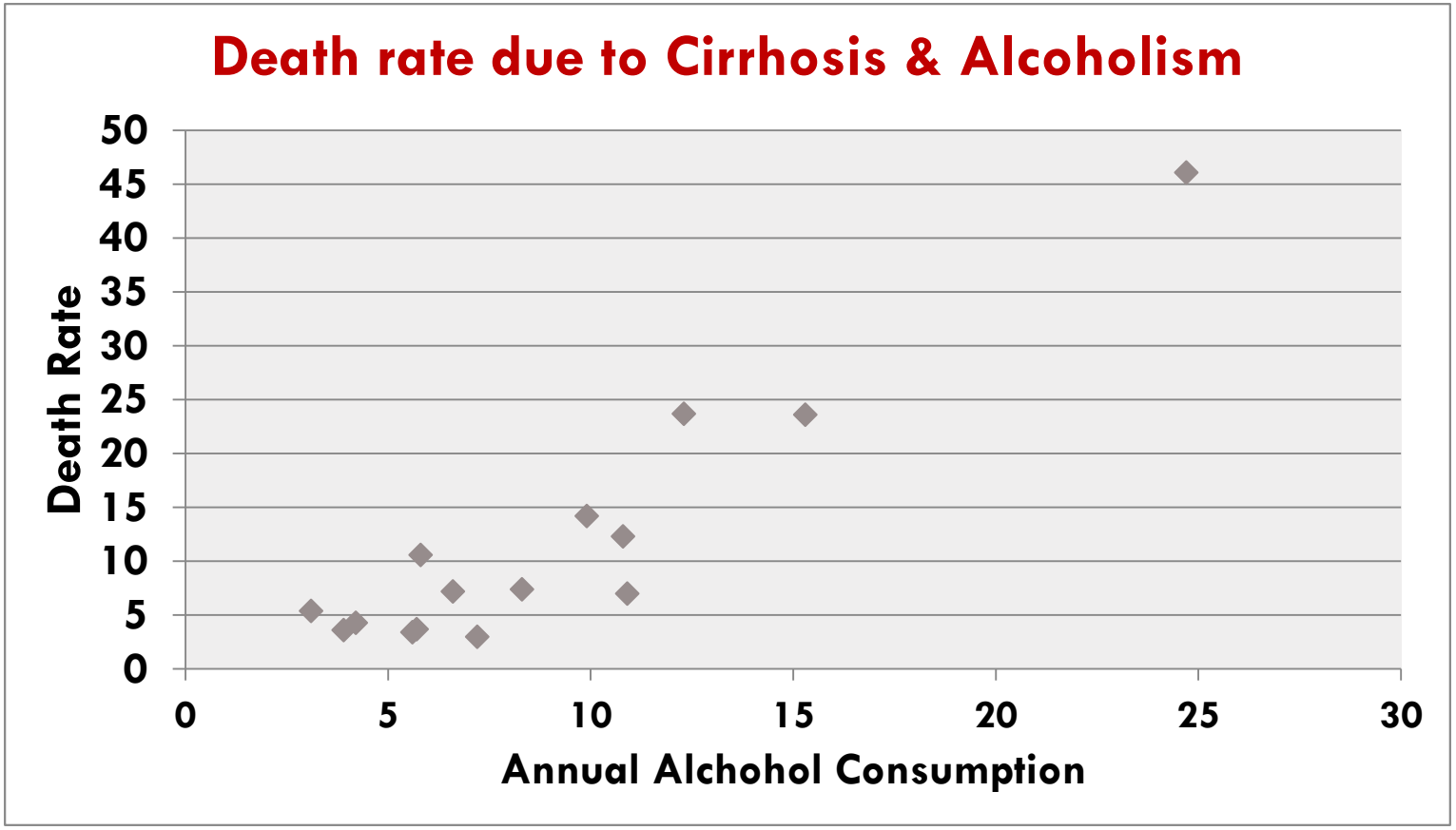
There are many problems that require a combined analysis of two variables.

Possible questions:

- Are these two variables related?
- How are these variables related?

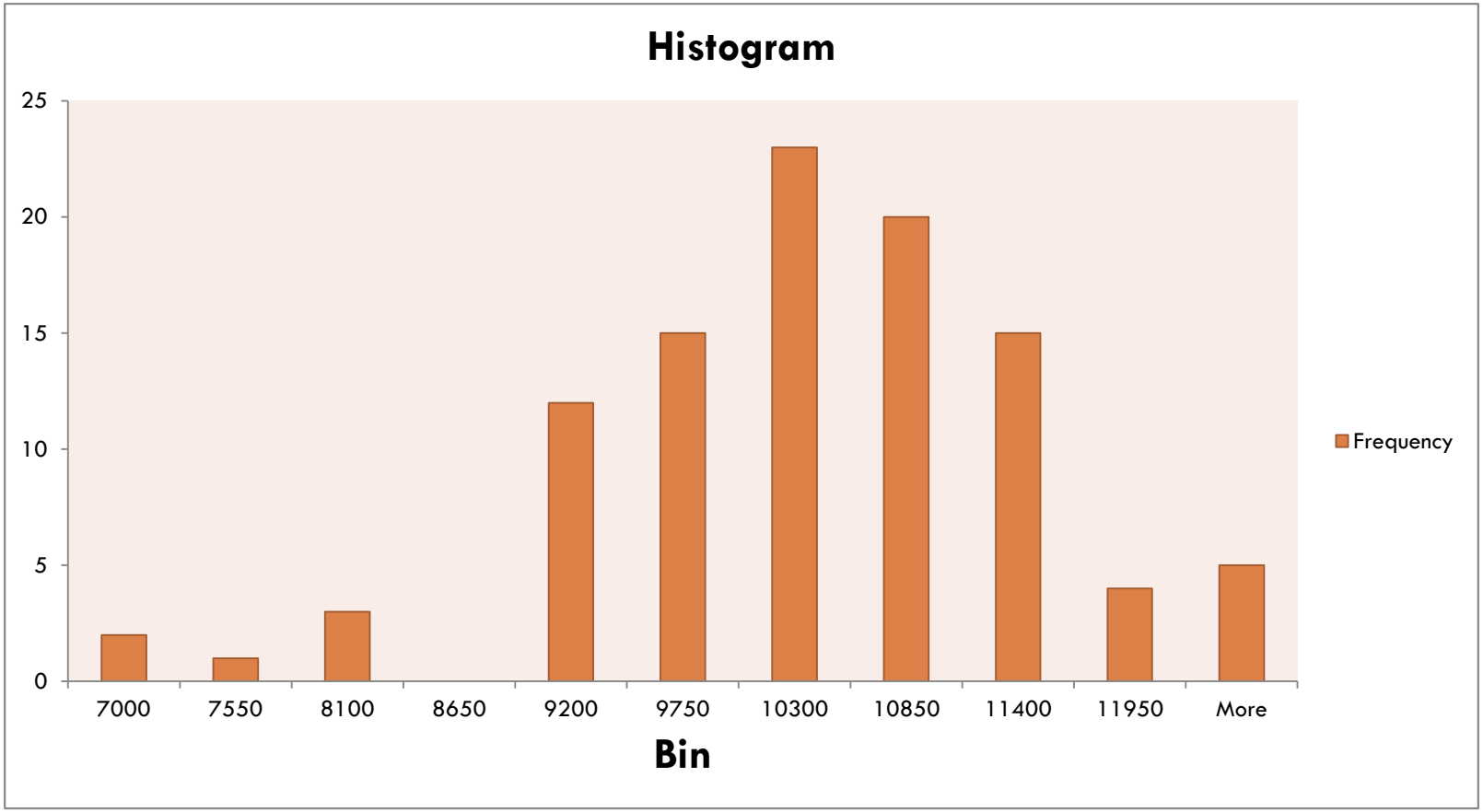
To answer such questions we plot these ordered pairs of bivariate data on a coordinate axis system. The independent variable are plotted on the horizontal axis and dependent variable are plotted on the vertical axis.

Example of a scatter plot:



Histogram

A histogram is a graphical display based on the frequency table.



Histogram

- **A histogram is a graphical display based on a frequency table.**
- In a histogram the frequencies are associated to numerical ranges called bins.
- To each bin is associated a bar whose area is proportional to the bin width.

- **Excel can correctly handle only the case when all bins have same width.**

Creating a Frequency Table

- Open the "Histogram" worksheet
- Cells B3:B132 and C3:C132 gives monthly rate of return of GM and CSCO respectively.
- Cells E7:E15 have the bin boundaries: the bins are $x \leq -0.3$, $-0.3 < x \leq -0.2$, $-0.2 < x \leq -0.1$, … , $0.3 < x \leq 0.4$, $0.4 < x$.

Creating a Frequency Table: Method 1

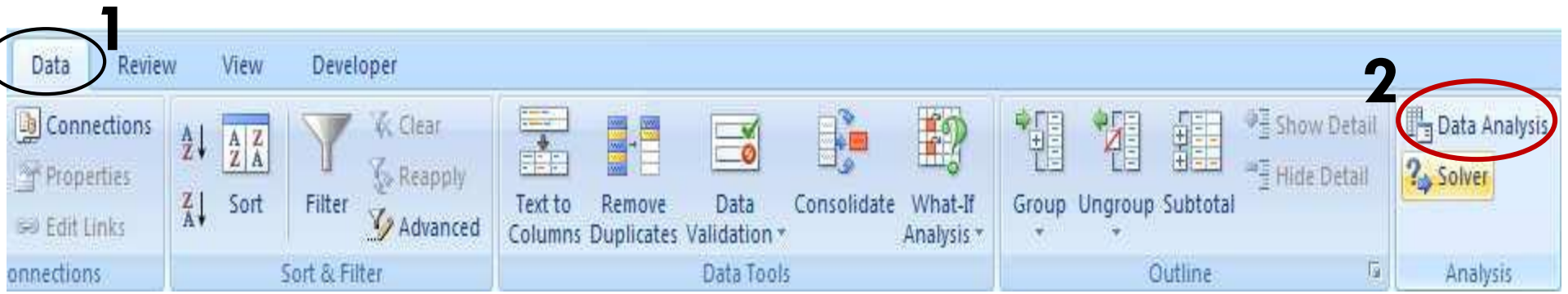
- Select cells F7:F15.
- Type "***=FREQUENCY(B3:B132, E7:E15)***".
- Press ***cntrl,shift,enter*** – all three keys together.
- In general, syntax of FREQUENCY is:

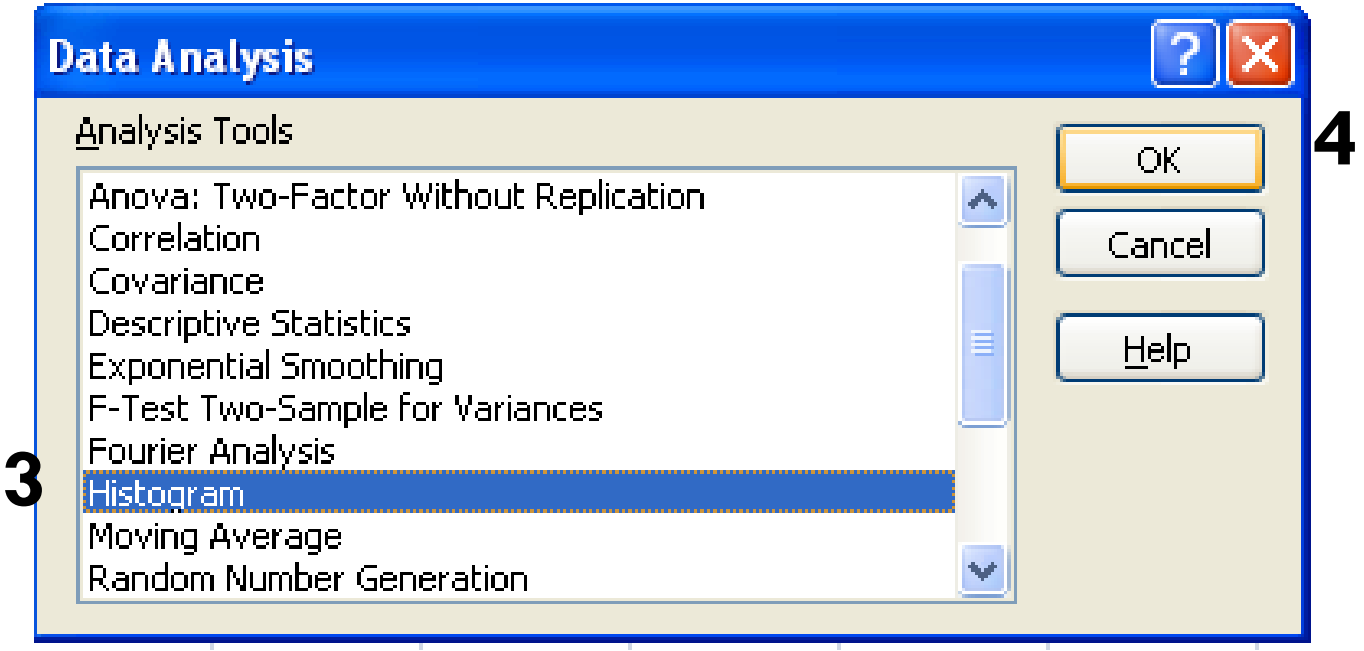
  FREQUENCY(data array, bin array) .

Creating a Frequency Table: Method 2

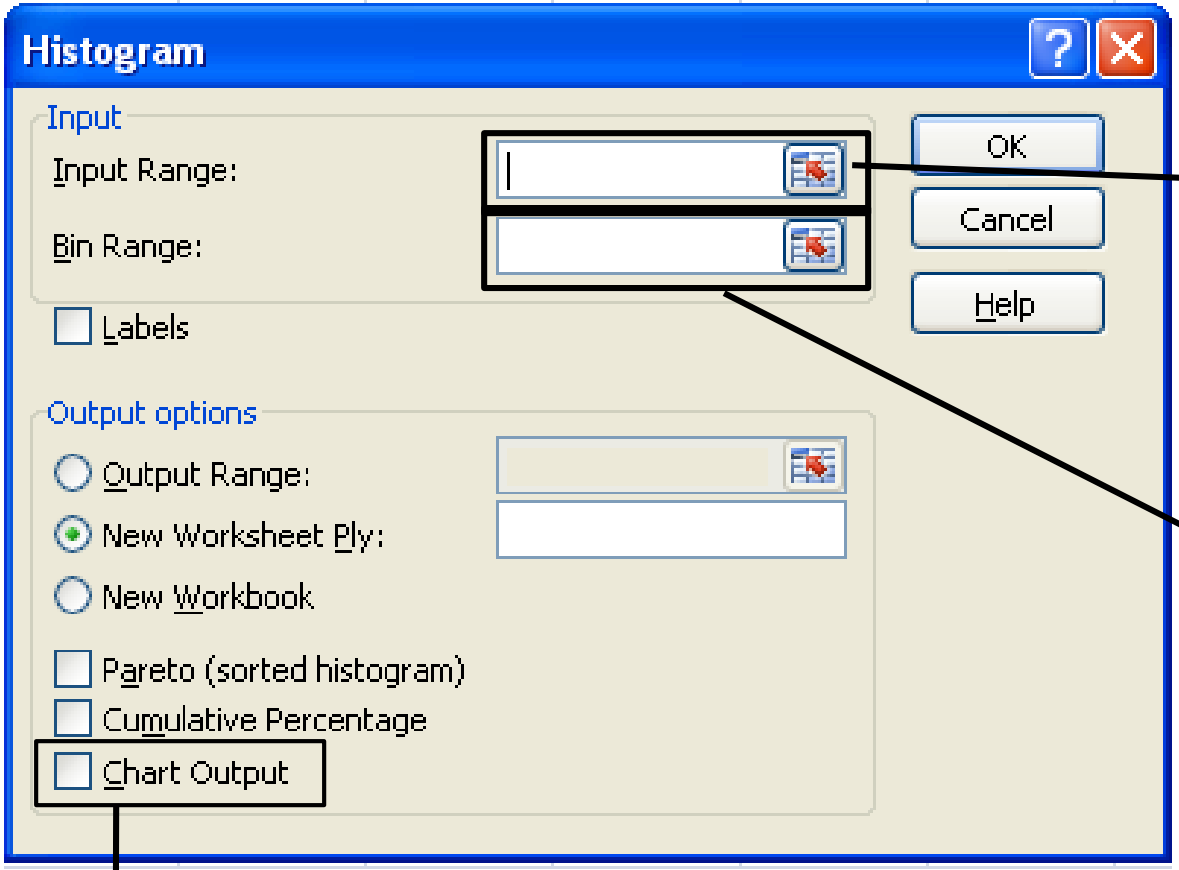
- In DATA tab, ANALYSIS group, select DATA ANALYSIS.
- In the Menu that opens, select HISTOGRAM
- Input Range is B3:C132, Bin Range is E7:E14
- In Output Options select “New Worksheet Ply” and “Chart Output”
- The frequency table and corresponding bar chart will be created in a new worksheet!

Histogram


1
Data
Review
View
Developer
2
Connections
Properties
Edit Links
onnections
Sort
Filter
Clear
Reapply
Advanced
Sort & Filter
Text to Columns
Remove Duplicates
Data Validation
Consolidate
What-If Analysis
Data Tools
Group
Ungroup
Subtotal
Show Detail
Hide Detail
Outline
Data Analysis
Solver
Analysis


Data Analysis
Analysis Tools
Anova: Two-Factor Without Replication
Correlation
Covariance
Descriptive Statistics
Exponential Smoothing
F-Test Two-Sample for Variances
Fourier Analysis
Histogram
Moving Average
Random Number Generation
3
OK
Cancel
Help
4

Histogram



Specify the range of the data

Specify the bin range. If left blank, excel divide the data into 10 bins.

Check this to get the chart

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